calculations in string theory improve on this mismatch and explain why the cosmological constant is zero, or if experiments do ultimately establish that its value is small but nonzero, can string theory provide an explanation? Should string theorists be able to rise to this challenge—as yet they have not—it would provide a compelling piece of evidence in support of the theory.

An Appraisal

The history of physics is filled with ideas that when first presented seemed completely untestable but, through various unforeseen developments, were ultimately brought within the realm of experimental verifiability. The notion that matter is made of atoms, Pauli's hypothesis that there are ghostly neutrino particles, and the possibility that the heavens are dotted with neutron stars and black holes are three prominent ideas of precisely this sort—ideas that we now embrace fully but that, at their inception, seemed more like musings of science fiction than aspects of science fact.

The motivation for introducing string theory is at least as compelling as any of these three ideas—in fact, string theory has been hailed as the most important and exciting development in theoretical physics since the discovery of quantum mechanics. This comparison is particularly apt because the history of quantum mechanics teaches us that revolutions in physics can easily take many decades to reach maturity. And compared to today's string theorists, the physicists working out quantum mechanics had a great advantage: Quantum mechanics, even when only partially formulated, could make direct contact with experimental results. Even so, it took close to 30 years for the logical structure of quantum mechanics to be worked out, and about another 20 years to incorporate special relativity fully into the theory. We are now incorporating general relativity, a far more challenging task, and, moreover, one that makes contact with experiment much more difficult. Unlike those who worked out quantum theory, today's string theorists do not have the shining light of nature—through detailed experimental results—to guide them from one step to the next.

This means that it's conceivable that one or more generations of physicists will devote their lives to the investigation and development of string theory without getting a shred of experimental feedback. The